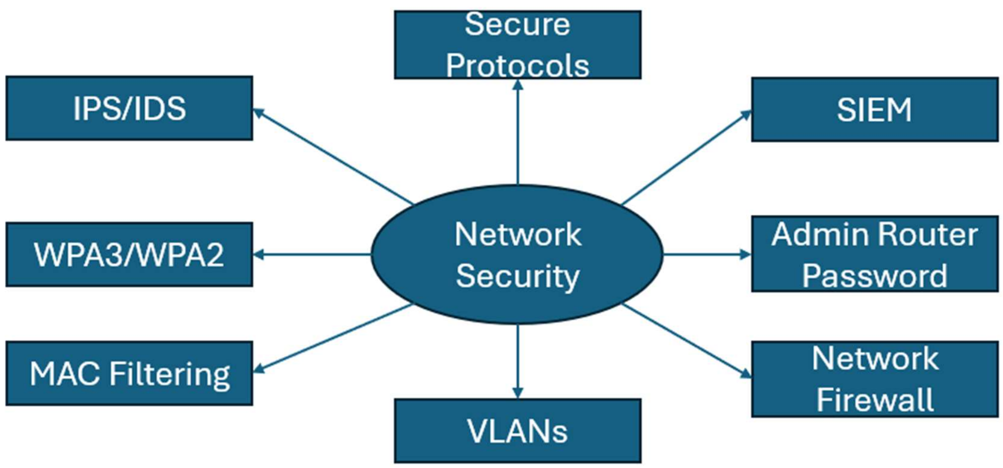
NETWORK SECURITY FRAMEWORK

Contents

Introduction

This network security framework is designed to help improve network security in home and company use cases. It was also designed to increase network security awareness as it is important users and companies protect their network from attack. This network security framework was developed to address the lack of depth into network security from my other frameworks. This was also developed as currently attackers can compromise systems due to known vulnerabilities in networks. This network security framework will cover basic protections that user can implement on their home network. It will also cover protections for companies, a secure network example and the trade-offs of this framework.

This framework cannot promise perfect network security and or perfect coverage over network protections. However, this framework can provide both basic network protections as well as some enterprise level protections. This framework should be used to create a secure network and should be adapted according to your needs and resources. This is because router models supporting different features and budget constrains can reduce security as certain tooling is unavailable.



The spider diagram above shows the main protections used in this network security framework. This diagram does not show the split between basic and advanced network-level protections. You can find more information in the sections below.

Basic Network Security Protections

- **It is important for home networks to have multiple layered security that raises the cost for an attacker while maintaining usability.**
- **There are 3 main protections that individuals can implement for their home network.**
- **The first protection is Wi-Fi Security Protocol**
- **Wi-Fi security protocols handle data security during transmit in wireless networks.**
- **Currently, there is WPA3 and WPA2 with personal and enterprise versions.**
- **I would recommend using WPA3 as it is the newest and most secure version but WPA2 works if you have no other choice.**
- **I would recommend never using WPA or WAP for the security protocol as they are vulnerable and have structural flaws.**
- **WPA3/WPA2-Personal is the recommended option as it requires one strong network password doesn't require any extra infrastructure to setup.**
- **WPA3-enterprise is the recommended option for companies and enterprise however it requires extra infrastructure and more management overhead.**
- **Wi-Fi security protocols prevent an attacker from reading the data travelling on a local network.**
- **The second protection is secure protocols**
- **Secure protocols are protocols that use encryption and other authentication methods to secure the transfer of data or connection between two devices.**
- **Secure protocols can include FTPS, SSH, SFTP and HTTPS on a network to ensure that network services are protected.**
- **I would avoid using insecure protocols like FTP, HTTP, SMB, RDP and UPnP.**
- **If you must use these insecure protocols, I would recommend using their most updated version and enforce authentication when possible.**
- **Secure protocols are important to ensure that network services cannot easily be exploited by an attacker**
- **The third protection is strong admin router password/network password.**
- **For WPA3-personal, you will need to ensure that the network password is somewhere between 16-24 characters.**
- **This is important to prevent brute-force attacks and to ensure that an attacker cannot easily read data on the network.**
- **The router should also have an admin password to prevent any physical tampering that an attacker may try.**
- **This admin password should be a 5–6-word passphrase for a mixture of memory and security.**
- **This is important as an attacker can tamper with network protocols, network traffic and certain devices with access to the router.**
- **I would also recommend turning off remote management and WPS on the router.**

Advanced Network Security Protections

- **It is important that company networks are secured with multiple layers but also usable and reliable for business operations.**
- **These advanced network security protections build on the basic security protections used for home networks.**
- **There are 4 main advanced network security protections.**
- **The first protection is VLANs.**
- **VLAN's are virtual local area networks used to segment traffic a network**
- **I would recommend using VLAN's based on different departments within the company for organisation and access control.**
- **VLAN's configuration and set-up can verify via router, and implementation can be complex and difficult to execute.**
- **VLAN's reduce the blast radius of compromise as an attacker cannot compromise the whole network from one VLAN.**
- **The second protection is IPS/IDS.**
- **IDS stands for intrusion detection system which is used to detect suspicious and malicious activity but this only alerts and doesn't prevent.**
- **IPS stands for intrusion prevention system which is used to prevent and stop malicious activity on a network.**
- **I would recommend layering both IDS and IPS as IDS allows for detection while IPS uses that detection to remove threats.**
- **IPS/IDS protect against attackers as they must bypass multiple detection and prevention systems to successfully compromise a network.**
- **The third protection is SIEM.**
- **SIEM stands for Security Information and Event Management which is a logging and monitoring tool that collects data from an IT environment.**
- **CSIRT and Security teams can use SIEM to detect, investigate and respond to security threats in real-time.**
- **However, SIEM can be complex to set-up and requires careful management and a dedicated security team to monitor logs.**
- **SIEM allows companies to respond to threats to real-time and investigate breaches however, I would only recommend SIEM for enterprises.**
- **The fourth protection is custom firewall rules**
- **Every network has a firewall which can be used to control network traffic inbound and outbound.**
- **I would recommend using whitelisting, blacklisting and blocking vulnerable ports that aren't used.**
- **Enterprises may use next-generation firewalls with AI-driven and real-time**
- **Other advanced protections include MAC filtering (Enterprise management) and NDR.**
- **MAC filtering is an advanced control as most users would struggle to implement.**

Home Network Example

- **This example will cover a multi layered security setup for a home network.**
- **First, the user is looking for simple network protections that can implement as an attacker recently compromised their data on their local network.**
- **Second, the user finds this network security framework and decides to implement the basic protections.**
- **Third, the user decides to start with the Wi-Fi security protocol protection.**
- **The user checks their router and notices they are using WPA2-personal with a weak network password.**
- **The user changes it to an WPA3-personal as they do not require the security of WPA3-enterprise and do not have the budget to set it up.**
- **The user sets a 16-character password stored in a password manager for the network to prevent brute force attacks.**
- **However, the user also must renter the network password or scan an QR code for every single device connected wirelessly to the network.**
- **Fourth, the user sets a strong admin password for their router.**
- **The user uses their password manager to generate a 5–6-word passphrase for their router.**
- **They may also enable MFA if their router supports however it is unlikely as most routers do not support this.**
- **This strong admin password prevents an attacker with physical access from tampering the router settings easily.**
- **Fifth, the user moves onto setting up secure protocols.**
- **The user looks at the services running on their network by using Nmap on their public facing IP. To be warned that Nmap is an technical tool and requires experience to use.**
- **The user notices from the scan that they use FTP, SMB and HTTP as vulnerable network services.**
- **The user configures their router to use FTPS or SFTP, to not use SMB and to use HTTPS instead of HTTP for traffic.**
- **This takes the user 2-3 hours as they must set-up Nmap, configure security protocols and troubleshoot any errors.**
- **Sixth, the user turns off remote management as they do not need for their home network and it is vulnerable to attacks.**
- **Seventh, the user also turns off WPS as they don't need it for any devices on their home network.**
- **The user is feeling more secure and hasn't had any major issues with network performance and reliability**
- **Remember that this is just an example and implementation will differ based on router model and the willingness of the user to be more secure.**
- **Some more high-risk users may implement MAC filtering at a basic level, IDS and VLAN's for guest networks.**

Network Security Framework Trade-Offs

- **This network security framework has trade-offs just like any other system I have built.**
- **There may also be further implementation challenges depending on budget, router model and technical knowledge.**
- **I have identified 5 main trade-offs with this framework.**
- **The first trade-off is cost.**
- **This trade-off exists as security tools especially ones like SIEM, IDS/IPS and using WPA3-enterprise can be extremely expensive.**
- **This means that the average user cannot afford to have the most advanced protections in terms of logging and monitoring capabilities.**
- **This trade-off can be managed by using built in tools, threat modelling to focus defences, and companies can allocate a specific budget for security if needed.**
- **The second trade-off is convenience.**
- **This trade-off exists as these protections require turning off WPS and not using vulnerable network protocols which were easier and more compatible to use.**
- **This means that security can be hard to implement as it isn't motivating and makes things more difficult in general.**
- **This trade-off can be managed by starting with implementing protections with low cost and accepting that security comes at the cost of convenience.**
- **The third trade-off is time-consumption.**
- **This trade-off exists as some security protections like using secure protocols, setting up SIEM, IDS/IPS, NDR and other tools can be time-consuming and take hours.**
- **This means that setting up network security for users can be difficult as people must go to work or school which means it's difficult to find time to set up security features.**
- **This trade-off can be managed by starting with the quickest security features, implementing features during weekends or holidays and working with others.**
- **The fourth trade-off is cognitive load**
- **This trade-off exists as it can be exhausting to be implementing multiple security features especially after work or school.**
- **This means that users can struggle to implement the features not because they can't do so but because they don't have the energy to do so.**
- **This trade-off can be managed by incremental progress, focusing on protections needed for your threat model and automating processes where possible.**
- **The fifth trade-off is compatibility**
- **This trade-off exists every router has different features meaning that some users can use WPA3 while some users can only use WPA2.**
- **This trade-off can be managed by using the second most secure option, built in security tools and external security tools if needed.**